CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 1 : Python Program to Solve 8-Puzzle Problem

Aim

To develop a Python program that solves the 8-Puzzle Problem by moving the blank tile from the initial state to reach the goal state.

Algorithm

- Represent the puzzle as a list of 9 elements (3×3 grid), with 0 as the blank tile.
- Define the initial state and the goal state.
- Display the initial state.
- Locate the blank tile (0) and slide it down by swapping it with the tile below.
- Locate the blank tile again and slide it right by swapping it with the tile on its right.
- Display the resulting state.
- Compare the resulting state with the goal state.
- If both are equal, print "Goal Reached"; otherwise print "Not Reached".

Program

```
puzzle = [1, 2, 3, 4, 0, 6, 7, 5, 8] goal
= [1, 2, 3, 4, 5, 6, 7, 8, 0]

def show(state):
print(state[:3])
print(state[3:6])
print(state[6:]) print("Initial
State:") show(puzzle)

# Move 1: slide the blank down b = puzzle.index(0)
puzzle[b], puzzle[b + 3] = puzzle[b + 3], puzzle[b]

# Move 2: slide the blank right b
= puzzle.index(0)
puzzle[b], puzzle[b + 1] = puzzle[b + 1], puzzle[b]

print("Goal State:") show(puzzle)

if puzzle == goal:
print("Goal Reached")
else:    print("Not
Reached")
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:/Users/shaik/OneDrive/Documents/1.py ================
Enter puzzle:  1 2 3 4 0 6 7 5 8
Enter goal:  1 2 3 4 0 6 7 5 8
Initial:
[1, 2, 3]
[4, 0, 6]
[7, 5, 8]
After move:
[1, 2, 3]
[4, 5, 6]
[7, 0, 8]
Not Reached
>>>
```

RESULT

Thus, the 8-Puzzle Problem was solved successfully using Python.